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Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
(ONLINE)
SEMESTER I
SESSION 2020/2021

COURSE NAME	:	OPERATING SYSTEM
COURSE CODE	:	BIC 20803
PROGRAMME CODE	:	BIS / BIP / BIW / BIM
EXAMINATION DATE	:	JANUARY / FEBRUARY 2021
DURATION	:	3 HOURS
INSTRUCTION	:	1. ANSWER ALL QUESTIONS 2. STUDENTS SHOULD UPLOAD ANSWER BOOKLET (PDF FORMAT) WITHIN 30 MINUTES AFTER EXAMINATION PERIOD.

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THIS QUESTION PAPER CONSISTS OF **FIVE (5)** PAGES

Q1 (a) Draw a tree diagram to illustrate the process creation in Figure **Q1(a)**.

```
int main()
{
        fork();
        fork();
        fork();
        printf ("Hello UTHM! \n");
}
```

FIGURE Q1(a)

(4 marks)

(b) Based on your answer in **Q1(a)**, write the output of the program.

(2 marks)

(c) Discuss **TWO (2)** effects if the parent process aborts the children processes.

(4 marks)

Q2 Table **Q2** shows list of processes (P1, P2, P3, P4 and P5) with their arrival and execution time. The time quantum is 6s (if required).

Process	Arrival Time (mm:ss)	Execution Time (s)
P1	00:00	18
P2	00:02	2
P3	00:08	5
P4	00:10	8
P5	00:11	28

TABLE Q2

Based on Table **Q2** above, answer the following questions.

(a) State **ONE (1)** name of scheduling algorithm that involved context switching to execute the processes in Table **Q2**.

(2 marks)

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(b) Based on your answer in **Q2(a)**,

(i) Draw a Gantt Chart to illustrate the execution process for all processes in Table **Q2**.

(4 marks)

(ii) Calculate the average waiting time and show your calculations.

(4 marks)

(iii) Calculate the average turnaround time and show your calculations.

(4 marks)

(c) Explain **TWO (2)** reasons that your scheduling answer in **Q2(a)** perform poorly compared to First Come, First Served (FCFS) scheduling. Draw a Gantt chart to support your answer.

(6 marks)

Q3 Figure **Q3** shows information about resources in a system.

- There are six classes of allocatable resouce labelled R1 through R6
- There are five processes labelled P1 through P5
- There are two instances of each resource
- There are some resource instances already allocated to processes, as follows:
  - One instance of R1 held by P1, another held by P2
  - One instance of R2 held by P3
  - One instance of R3 held by P3, another held by P4
  - One instance of R4 held by P3
  - One instance of R5 held by P3, another held by P4
  - Two instance of R6 held by P5
- Some processes have requested additional resources, as follow:
  - P1 wants one instance of R4
  - P2 wants one instance of R1 and one instance of R2
  - P3 wants one instance of R1
  - P4 wants one instance of R6
  - P5 wants one instance of R3

FIGURE Q3

(a) Draw a resource allocation graph to illustrate the information in Figure **Q3**.

(10 marks)

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(b) Based on your answer in **Q3(a)**, analyze state for each process.

(5 marks)

(c) List **FIVE (5)** possible sequences to avoid deadlock.

(5 marks)

Q4 Figure **Q4** shows one type of level directory.

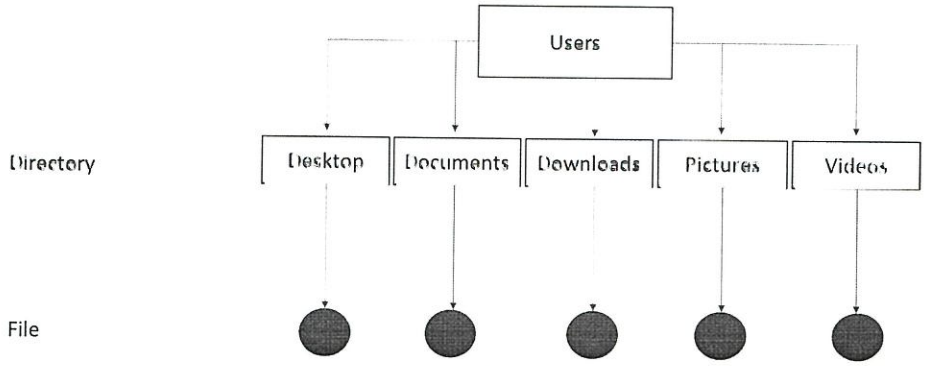

Figure Q4

(a) List **TWO (2)** disadvantages of having type of level directory in the figure.

(4 marks)

(b) Illustrate a two-level directory diagram if your laptop has two users with same directories and files.

(6 marks)

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Q5 (a) Figure **Q5** shows jobs allocation in a main memory by using Best-Fit Allocation Scheme.

Job No.	Memory Requested	Status
Job 1	15K	Allocated at Block 4
Job 2	50K	Allocated at Block 3
Job 3	350K	wait
Job 4	200K	Allocated at Block 1
Job 5	110K	Allocated at Block 5

Memory Block No.	Memory Block Size	Job No.	Job Size	Status	Total Waste
Block 1	300K	Job 4	200K	Busy	100
Block 2	250K			Free	
Block 3	50K	Job 2	90K	Busy	0
Block 4	20K	Job 1	15K	Busy	5
Block 5	115K	Job 5	110K	Busy	5

FIGURE Q5

Draw and label the positions of external and internal fragmentations.

(4 marks)

(b) Assume a main memory is composed of two-page frames for public use and that a program requests pages in the following order:

A, B, D, D, D, B, A, A, B, C, C, B

Using the First-In-First Out (FIFO) page removal algorithm, perform a page trace analysis indicating page faults with asterisks (*).

(6 marks)

-END OF QUESTIONS-

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